HYDRANAL™Composite 5

34805-1L

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1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

Product information

Trade name	:	HYDRANAL™Composite 5
Number	:	000000020610
Recommended use of the chemical and restrictions on use	:	Laboratory chemicals
Manufacturer or supplier's details	:	Honeywell Specialty Chemicals Seelze GmbH Wunstorfer Straße 40 Seelze, 30926
For further information, please contact:	:	1-800-368-0050 +1-231-726-3171 (Monday-Friday, 9:00am-5:00pm)
In case of emergency call	:	**Medical: 1-800-498-5701 or +1-303-389-1414**
	:	**Transportation (CHEMTREC): 1-800-424-9300 or +1-703-527-3887**
		In Vietnam: +(84)-838012436
	:	(24 hours/day, 7 days/week)

2. HAZARDS IDENTIFICATION

Classification of the substance or mixture

Classification of the substance or mixture : Acute toxicity, Category 5, Oral
Acute toxicity, Category 5, Dermal
Acute toxicity, Category 5, Inhalation
Carcinogenicity, Category 2
Reproductive toxicity, Category 1B
Specific target organ toxicity - repeated exposure, Category 1
Short-term (acute) aquatic hazard, Category 2

GHS Label elements, including precautionary statements

Symbol(s) :



Signal word : Danger

Hazard statements	:	May be harmful if swallowed. May be harmful in contact with skin. May be harmful if inhaled. Suspected of causing cancer. May damage fertility or the unborn child. Causes damage to organs through prolonged or repeated exposure. Toxic to aquatic life.
Precautionary statements	:	**Prevention:** Obtain special instructions before use. Do not handle until all safety precautions have been read and understood. Do not breathe dust/ fume/ gas/ mist/ vapours/ spray. Wash skin thoroughly after handling. Do not eat, drink or smoke when using this product. Avoid release to the environment. Use personal protective equipment as required. **Response:** IF INHALED: Call a POISON CENTER/ doctor if you feel unwell. IF exposed or concerned: Get medical advice/ attention. **Storage:** Store locked up. **Disposal:** Dispose of contents/ container to an approved waste disposal plant.

3. COMPOSITION/INFORMATION ON INGREDIENTS

Chemical nature : Mixture

Chemical name	CAS-No.	Concentration
2-(2-Ethoxyethoxy)ethanol 2-(2-Ethoxyethoxy)ethanol	111-90-0	>65.00 - <75.00 %
Imidazole Imidazole	288-32-4	>=5.00 - <10.00 %
Iodine Iodine	7553-56-2	>=5.00 - <10.00 %
Sulphur dioxide	7446-09-5	>=5.00 - <10.00 %

Sulphur dioxide		
Hydriodic acid Hydriodic acid	10034-85-2	>=5.00 - <10.00 %
2-Methylimidazole 2-Methylimidazole	693-98-1	>=5.00 - <10.00 %

4. FIRST AID MEASURES

General advice	:	First aider needs to protect himself. Move out of dangerous area. Take off all contaminated clothing immediately.
Inhalation	:	Move to fresh air. Keep patient warm and at rest. Call a physician immediately.
Skin contact	:	Wash off immediately with plenty of water. If skin irritation persists, call a physician.
Eye contact	:	In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice. Protect unharmed eye.
Ingestion	:	When swallowed, allow water to be drunk. Do NOT induce vomiting. Call a physician immediately.
Notes to physician	:	Treat symptomatically.

5. FIREFIGHTING MEASURES

Suitable extinguishing media	:	Water spray Foam Carbon dioxide (CO2) Dry powder
Unsuitable extinguishing media	:	Do not use a solid water stream as it may scatter and spread fire.
Specific hazards during firefighting	:	Fire may cause evolution of: Carbon monoxide Carbon dioxide (CO2) Sulphur oxides nitrogen oxides (NOx)

Special protective equipment for firefighters	:	Wear an approved positive pressure self-contained breathing apparatus in addition to standard fire fighting gear.
Further information	:	Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

6. ACCIDENTAL RELEASE MEASURES

Personal precautions, protective equipment and emergency procedures	:	Evacuate personnel to safe areas. Wear personal protective equipment. Unprotected persons must be kept away. Ensure adequate ventilation. Remove all sources of ignition.
Environmental precautions	:	Should not be released into the environment.
Methods and materials for containment and cleaning up	:	Ventilate the area. Contain spillage, soak up with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and transfer to a container for disposal according to local / national regulations (see section 13).

7. HANDLING AND STORAGE

Handling

Precautions for safe handling	:	Wear personal protective equipment. Use only in well-ventilated areas.
Advice on protection against fire and explosion	:	Keep away from sources of ignition - No smoking. Normal measures for preventive fire protection.

Storage

Conditions for safe storage, including any incompatibilities	:	Keep only in the original container, tightly closed, in a well ventilated place. Store at room temperature. (Ambient temperature: > 0 < 35°C) Protect from atmospheric moisture and water.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Components with workplace control parameters

Components	CAS-No.	Value	Control parameters	Update	Basis
Iodine Iodine	7553-56-2	TWA : Time weighted average (TWA):	1 mg/m3	10 2002	VN OEL:Vietnam. OELs, as amended
		STEL : Short-term exposure limit (STEL):	2 mg/m3	10 2002	VN OEL:Vietnam. OELs, as amended
		STEL : Short-term exposure limit (STEL):	2 mg/m3	10 2002	VN OEL:Vietnam. OELs, as amended
		TWA : Time weighted average (TWA):	1 mg/m3	10 2002	VN OEL:Vietnam. OELs, as amended
Sulphur dioxide Sulphur dioxide	7446-09-5	TWA : Time weighted average (TWA):	5 mg/m3	10 2002	VN OEL:Vietnam. OELs, as amended
		STEL : Short-term exposure limit (STEL):	10 mg/m3	10 2002	VN OEL:Vietnam. OELs, as amended

Appropriate engineering controls

Use with local exhaust ventilation.
Prevent vapour buildup by providing adequate ventilation during and after use.

Individual protection measures, such as personal protective equipment

Respiratory protection : In case of insufficient ventilation wear suitable respiratory equipment.

Hand protection	:	Wear nitrile rubber gloves to avoid contact with the skin. Gloves must be inspected prior to use. Replace when worn.
Eye protection	:	Safety goggles
Skin and body protection	:	Protective suit
Hygiene measures	:	Take off all contaminated clothing immediately. Remove and wash contaminated clothing before re-use. Wash hands before breaks and at the end of workday. When using do not eat or drink.
Protective measures	:	Ensure that eyewash stations and safety showers are close to the workstation location. Legal requirements are to be considered in regard of the selection, use and care of personal protective equipment. Avoid exposure - obtain special instructions before use.

9. PHYSICAL AND CHEMICAL PROPERTIES

Physical state	:	liquid
Colour	:	brown
Odour	:	characteristic
pH	:	4.5 - 5.5 at , 20 °C
Melting point/range	:	Note: no data available
Boiling point/boiling range	:	194 °C at 1,013 hPa
Flash point	:	210 °F (99 °C)
Evaporation rate	:	Note: no data available
Flammability	:	Not applicable
Lower explosion limit	:	Note: no data available

Upper explosion limit	:	Note: no data available
Vapour pressure	:	Note: no data available
Vapour density	:	Note: no data available
Density	:	ca. 1.17 g/cm3 at 25 °C
Water solubility	:	Note: completely miscible
Partition coefficient: n-octanol/water	:	Note: no data available
Ignition temperature	:	Note: no data available
Decomposition temperature	:	Note: No decomposition if used as directed.
Viscosity, dynamic	:	Note: no data available
Viscosity, kinematic	:	Note: no data available
Oxidizing properties	:	The substance or mixture is not classified as oxidizing.

10. STABILITY AND REACTIVITY

Chemical stability	:	Stable under normal conditions.
Possibility of hazardous reactions	:	Hazardous polymerisation does not occur.
Conditions to avoid	:	Protect from atmospheric moisture and water. Heat, flames and sparks.
Incompatible materials to avoid	:	Strong oxidizing agents
Hazardous decomposition products	:	In case of fire hazardous decomposition products may be produced such as: Carbon monoxide

Carbon dioxide (CO2)
Sulphur oxides
nitrogen oxides (NOx)

11. TOXICOLOGICAL INFORMATION

Acute oral toxicity	:	LD50: > 2,000 mg/kg Species: Rat Method: OECD 423
Acute inhalation toxicity	:	Note: no data available
Acute dermal toxicity	:	LD50: > 2,000 mg/kg Species: Rat Method: OECD Test Guideline 402
Skin irritation	:	Species: Rabbit Result: No skin irritation Method: OECD Test Guideline 404
Eye irritation	:	Species: Rabbit Result: No eye irritation Method: OECD Test Guideline 405
Sensitisation		
2-(2-Ethoxyethoxy)ethanol	:	Species: Guinea pig Classification: non-sensitizing
2-Methylimidazole	:	Mouse local lymph node assay Species: Mouse Result: Did not cause sensitisation on laboratory animals. Method: OECD Test Guideline 429
Genotoxicity in vitro	:	Test Method: Ames test Result: negative
Genotoxicity in vivo		
2-(2-Ethoxyethoxy)ethanol	:	Species: Rat, male Cell type: Liver cells

Application Route: Oral
Method: OECD Test Guideline 486
Result: negative

: Test Method: Chromosome aberration test
Species: Mouse, male
Cell type: Bone marrow
Application Route: Intraperitoneal injection
Method: OECD Test Guideline 474
Result: negative

Imidazole : Test Method: Micronucleus test
Species: Mouse, male and female
Cell type: Bone marrow
Application Route: Oral
Method: OECD Test Guideline 474
Result: negative

Teratogenicity
Imidazole : Species: RatApplication Route: Oral

No observed adverse effect level: 60 mg/kg body weight
No observed adverse effect level: 60 mg/kg body weight
Method: OECD Test Guideline 414
Result: Embryotoxic effects and adverse effects on the offspring were detected.

2-Methylimidazole : Species: RatApplication Route: Oral

No observed adverse effect level: >50 mg/kg body weight
No observed adverse effect level: 2 mg/kg body weight
Method: OECD Test Guideline 414
Result: Embryotoxic effects and adverse effects on the offspring were detected.

Further information : Note: Causes damage to organs through prolonged or repeated exposure (Thyroid).

12. ECOLOGICAL INFORMATION

Toxicity to fish
2-(2-Ethoxyethoxy)ethanol : flow-through test
LC50: 6,010 mg/l

Exposure time: 96 h
Species: Ictalurus punctatus (channel catfish)
Method: OECD Test Guideline 203

Imidazole : static test
LC50: 283.6 mg/l
Exposure time: 48 h
Species: Leuciscus idus (Golden orfe)

Iodine : LC50: 1.67 mg/l
Exposure time: 96 h
Species: Oncorhynchus mykiss (rainbow trout)

2-Methylimidazole : static test
LC50: 190 mg/l
Exposure time: 96 h
Species: Leuciscus idus (Golden orfe)
Method: DIN 38412

Toxicity to daphnia and other aquatic invertebrates

2-(2-Ethoxyethoxy)ethanol : static test
LC50: 1,982 mg/l
Exposure time: 48 h
Species: Daphnia magna (Water flea)
Method: OECD Test Guideline 202

Imidazole : static test
EC50: 341.5 mg/l
Exposure time: 48 h
Species: Daphnia magna (Water flea)
Method: Directive 67/548/EEC, Annex V, C.2.

Iodine : LC50: 0.55 mg/l
Exposure time: 48 h
Species: Daphnia magna (Water flea)

2-Methylimidazole : static test
EC50: 225.31 mg/l
Exposure time: 48 h
Species: Daphnia magna (Water flea)
Method: Directive 67/548/EEC, Annex V, C.2.

Toxicity to algae

Imidazole : static test
EC50: 133 mg/l
Exposure time: 72 h
Species: Desmodesmus subspicatus (green algae)

Method: DIN 38412

Iodine	:	Growth inhibition EC50: 0.13 mg/l Exposure time: 72 h Species: Desmodesmus subspicatus (green algae) Method: OECD Test Guideline 201
2-Methylimidazole	:	static test EC50: 256.3 mg/l Exposure time: 72 h Species: Desmodesmus subspicatus (Scenedesmus subspicatus) Method: DIN 38412 static test EC50: 189 mg/l Exposure time: 72 h Species: Desmodesmus subspicatus (Scenedesmus subspicatus) Method: DIN 38412

Persistence and degradability

Biodegradability	:	Result: Readily biodegradable. Value: 78 % Method: OECD 302 B

Additional ecological information

Iodine	:	Harmful to aquatic organisms, may cause long-term adverse effects in the aquatic environment.

13. DISPOSAL CONSIDERATIONS

Disposal methods	:	In accordance with local and national regulations.

14. TRANSPORT INFORMATION

ADR
Not dangerous goods

IATA
Not dangerous goods

IMDG
Not dangerous goods

RID
Not dangerous goods

15. REGULATORY INFORMATION

National regulatory information

: No information available.

Other international regulations

Notification status

US. Toxic Substances Control Act	:	All chemical substances in this product are either listed on the TSCA Inventory or are in compliance with a TSCA Inventory exemption.
Australia. Industrial Chemical (Notification and Assessment) Act	:	On the inventory, or in compliance with the inventory
China. Inventory of Existing Chemical Substances (IECSC)	:	On the inventory, or in compliance with the inventory
Note	:	Note: Because of the potential specific inventory listing of components of this product line, further, more detailed information can be requested from SafetyDataSheet@Honeywell.com.

16. OTHER INFORMATION

		HMIS III	NFPA
Health hazard	:	2	2
Flammability	:	1	1
Physical Hazard	:	0	
Instability	:		0

Hazard rating and rating systems (e.g. HMIS® III, NFPA): This information is intended solely for the use of individuals trained in the particular system.

Further information

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text. Final determination of suitability of any material is the sole responsibility of the user. This information should not constitute a guarantee for any specific product properties.

Changes since the last version are highlighted in the margin. This version replaces all previous versions.
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